

Do Retention Elections Create Cycles in Criminal Sentencing?

An Out-of-Sample Replication

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Abstract

Judicial retention elections are low-salience, uncompetitive events where incumbents virtually never lose. However, some articles have found that criminal court judges become significantly more punitive as these safe elections approach. Using newly available data spanning multiple election cycles, we conduct an out-of-sample replication of the most widely cited article in this literature ([Huber and Gordon 2004](#)) and find no evidence of election cycle effects.

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Introduction

Judicial elections are a uniquely American phenomenon. Trial court judges — who hear the vast majority of criminal and civil cases — are selected or retained via election in 38 of the 50 states ([Kulleen 2024](#)). A significant literature bridging political science, economics, and legal studies has grown up around the ramifications of judicial elections, highlighting the tension between accountability and fairness in the courts. In particular, an ongoing debate asks whether elected judges hand down harsher sentences under electoral pressure. This literature has experienced something of a revival in recent years, as a new generation of scholars probe the causes and extent of electoral influence on judicial decision-making ([Dippel and Poyker 2021](#); [Lu 2024](#); [Boston and Silveira 2024](#)).

The most significant article in this vein of research is [Huber and Gordon \(2004\)](#)’s study of election cycle effects in Pennsylvania trial courts. They find that judges facing retention elections grew more punitive as election day approached — amounting to two thousand additional years of incarceration for those convicted during the nine years they studied. Such a massive punitive shift is theoretically puzzling given the institutional safety of trial court retention elections. For context, from 2001 to 2023, only one Pennsylvania Court of Common Pleas judge lost a retention election.¹ Voter information is also very low, even following the advent of “new style” judicial campaigns ([Greenberg Quinlan Rosner Research 2001](#)). It is surprising that an uncompetitive and low-information environment would generate such drastic behavioral changes on the bench.

This article has become a staple of the judicial politics literature, a go-to citation when discussing the effect of elections on criminal justice outcomes. Scholars cite [Huber and Gordon \(2004\)](#) both generally as evidence of judicial responsiveness to public

¹Blair County Judge Elizabeth Doyle lost a retention election in 2023. Her loss was precipitated by decisions regarding child custody agreements during the COVID lockdown and bail reductions, rather than criminal sentencing ([Stephens 2023](#)).

opinion, and in particular evidence of cyclical variation in that responsiveness. For example, the article has been cited as evidence that “judges’ decisions... are influenced by impending retention elections or reappointment” (Rachlinski and Wistrich 2017) and “that judges... worry about reelections and become harsher when they loom” (Tonry 2024). This article’s influence is felt beyond the scope of judicial politics as well — Huber and Gordon (2004) has been cited in studies of state policymaking (Lax and Phillips 2009), education politics (Hess 2008), business ethics (Mansouri and Roney 2014), public goods provision in Uganda (Grossman and Baldassarri 2012), investigations into members of the Italian parliament (Ceron and Mainenti 2015), immigration enforcement (King et al. 2012), and local tax policy (Jensen et al. 2015).

In this paper, we aim to determine whether these findings of electoral cycle effects in sentencing remain robust. The time elapsed since the publication of Huber and Gordon (2004)’s influential article provides a unique opportunity to test their hypotheses using out-of-sample data. Leveraging twenty years of new data from the Pennsylvania Court of Common Pleas, we find no evidence of election cycles in sentencing. Judges’ punitiveness does not increase toward the end of their term. It now appears that Pennsylvania’s 10-year retention election system does not generate election-driven punitive cycles. We also find that the cycle effect is bounded strictly within the 1990–1994 window, suggesting that the original results may be an idiosyncratic anomaly of that specific era.

Judicial Election Cycles

Voters are said to be myopic, weighing recent information and experiences more heavily when making voting decisions (Achen and Bartels 2016). Voter myopia contributes to the theory of the electoral business cycle, whereby elected representatives pull the levers of macroeconomic policy to boost voter evaluations of the economy close to election day (Nordhaus (1975), Tufte (1978), see review in Franzese (2002)). While presidents and representatives may be judged by the state of the economy, other elected officials may

be held to different metrics of success. [Huber and Gordon \(2004\)](#) brought judges into the electoral cycle literature by suggesting that judges can signal their quality to voters through sentencing. Judges are primarily concerned with triggering “fire alarm” oversight, which they argue “almost always correspond to perceived instances of *underpunishment*, not overpunishment” (p. 249). When making sentencing decisions, judges “continuously reevaluate the balance between the value of office and implementing their own preferences” (p. 250). This balance shifts closer to election day, when the authors argue voters are making evaluations of judges and don’t have time to forget negative news coverage. Driven by the desire to avoid negative media attention for their sentencing decisions, the authors argue that “judges will become more punitive... over the course of their terms” (p. 250).

However, the logic underlying the cycle effect is difficult to square with the reality of judicial retention elections. In addition to judges facing an almost certain chance of re-election, voter knowledge in these races is extremely low. Only about half of voters know if the justices on their state Supreme Court are elected or not ([National Center for State Courts 2009](#)). Supermajorities do not recall hearing anything about judicial races and cannot name a candidate or the incumbent when asked ([Johnson et al. 1978](#); [McKenzie 2012](#)). While the “new style” judicial campaigns since the 1990s have weighed in on high-salience issues, voter knowledge remains low ([Hojnacki and Baum 1992](#); [Greenberg Quinlan Rosner Research 2001](#); [Gibson 2009](#)). Of course, elected officials may represent constituent preferences even when the public is inattentive, either because their preferences are aligned with the public’s or to minimize the risk of drawing the attention of interested groups and individuals who could jeopardize their re-election efforts ([Miller and Stokes 1963](#)). [Huber and Gordon \(2004\)](#) argue that the lack of information about judicial elections actually contributes to the fear of fire-alarm oversight. They argue that, “[u]nder conditions of near absolute voter ignorance, information about the adverse consequences of a single case, when publicized, can be decisive in swaying voter opinion against a presiding judge” (p. 249).

Recent developments in the politics of punishment also raise questions about whether effects observed in past decades persist. The existing work on electoral cycles assumes that the median voter wants harsher sentencing. [Berdejó and Yuchtman \(2013\)](#) address this assumption directly, pointing to the fact that 82.8% of respondents to the General Social Survey between 1972 and 2006 believed courts were “not harsh enough” on criminals (p. 743). However, attitudes towards criminal justice have evolved over the last decade. As of 2024, the share of adults thinking courts were too lenient had fallen to 58% [Davern et al. \(2024\)](#). In another survey from 2021, 28% of national survey respondents believed people spent too long in prison, compared to 32% who responded that sentences are not long enough ([Gramlich 2021](#)). Unsurprisingly, the evolution in attitudes towards sentencing has been especially notable among liberals. If judges under electoral pressure change their behavior to appeal to the median voter, it is not clear that they should necessarily become more punitive. In liberal jurisdictions, the median voter may prefer sentences to stay the same or even become more lenient.

In their foundational article on judicial elections in Pennsylvania, [Huber and Gordon \(2004\)](#) find evidence that judges facing retention elections sentence more harshly close to election day than they do at the beginning of their term. This phenomenon was confirmed again in [Gordon and Huber \(2007\)](#)’s study of Kansas courts, though the effect seems to hold in districts that employ competitive, partisan elections to retain judges, and not in the retention election districts (see [Lim \(2013\)](#) for partial confirmation of these results). Since the publication of these two articles, election cycle effects were also found in Washington ([Berdejó and Yuchtman 2013](#)) and North Carolina ([Dippel and Poyker 2021](#)) — both states with competitive elections. [Dippel and Poyker \(2021\)](#) do not find effects in Colorado, the other state in their sample where judges face retention elections. Further, [Lu \(2024\)](#) narrows the scope of analysis to one county (Cook County, IL) and uses the randomized assignment of cases to judges to aid in causal identification. Their results yield no evidence of cycle effects in sentencing.

As a part of a ten-state study of cycle effects in sentencing, [Dippel and Poyker](#)

(2021) perform an out-of-sample replication of [Huber and Gordon \(2004\)](#). Analyzing sentencing in Pennsylvania from 2001-2016, they find extremely large election cycle effects — approximately 15 additional months of incarceration in cases under maximal as opposed to minimal electoral pressure. However, their results are somewhat ambiguous and contradictory. While their point estimate for the election cycle effect is large, so are their standard errors. Puzzlingly, the effect is only significant with robust, rather than conventional, standard errors. Their sample size is also much smaller than our own, despite including a greater number of offenses.² As their replication data are unavailable (likely due to state restrictions on the sharing of sentencing data), we are unable to replicate their results to account for these differences. However, our sample encompasses theirs and adds five additional years of sentencing data from 2017-2021.

Given the mixed empirical record, establishing the boundary conditions of judicial election cycles remains a critical task. The most striking anomaly in this literature is the persistent claim that uncompetitive, low-salience retention elections generate severe punitive cycles ([Huber and Gordon 2004](#)). Leveraging two decades of out-of-sample data from the Pennsylvania Court of Common Pleas, we test whether these original findings hold in an institutional environment characterized by near-universal re-election and minimal voter knowledge.

Data and Methods

We obtained case-level sentencing data from the Pennsylvania Court of Common Pleas.³ Common Pleas judges are first selected through partisan, competitive elections. Once a judge has won a seat, they face a non-competitive retention election after each ten-year term. By following each judge over the course of multiple terms, we assemble a panel of

²[Dippel and Poyker \(2021\)](#) use the case selection criteria from [Berdejó and Yuchtman \(2013\)](#), which includes murder and manslaughter alongside the offenses in [Huber and Gordon \(2004\)](#) (aggravated assault, rape, robbery). [Dippel and Poyker \(2021\)](#) report 12,866 cases from 2001-2016. For that time period and offense selection, we have 45,501 cases.

³While our replication code is public, our data agreement with the Pennsylvania Commission on Sentencing prohibits us from sharing the sentencing data. Our replication files include the sufficient statistics necessary to replicate our linear estimates.

their decision-making across varying periods of electoral pressure. [Huber and Gordon \(2004\)](#) employ nine years of sentencing data from the Pennsylvania Court of Common Pleas, from 1990 to 1999. These data are not allowed to be shared publicly, so we made a separate data request to the Pennsylvania Commission on Sentencing for all Court of Common Pleas cases with sentencing dates from 1990 to 2021. When we analyze the data from the same period as [Huber and Gordon \(2004\)](#), we obtain similar results (Tables [A.5](#) and [A.6](#) in the Appendix). We are also able to combine the three decades of data for pooled analysis. One advantage of our out-of-sample data is that we are able to observe multiple election cycles — as many as three separate terms (two elections) per judge. In the pooled analysis, we can observe as many as three elections per judge. This allows us to analyze the data with judge fixed effects, isolating within-judge effects that are identified by over-time variation in election proximity.

First, we re-create the model specifications from the main results in [Huber and Gordon \(2004\)](#). Though [Huber and Gordon \(2004\)](#) employ two-limit tobit models (which we replicate with similar results in Table [A.6](#)), our main analysis follows more recent work in using linear regression models ([Berdejó and Yuchtman 2013](#); [Dippel and Poyker 2021](#)).⁴ When someone is sentenced in Pennsylvania, they receive both a minimum and a maximum sentence. Following Huber and Gordon, the outcome variable is the minimum sentence, top-coded at 720 months (including life sentences and death sentences).

In each model, the explanatory variable *Proximity* is a linear measure of distance to the next election, ranging from 0 to 1. To determine the proximity to the sentencing judge’s next election, we matched each case in the sentencing records with the relevant election records. A proximity score of 0 is equivalent to the first day of a judge’s term, while a proximity score of 1 is the last day of the term, ten years later.

Identification of electoral cycle effects relies on the assumption that the “treatment” of electoral proximity is quasi-random with respect to case characteristics. Table [1](#)

⁴Huber and Gordon report that “[f]ixed-effects OLS... produces nearly identical results” (p. 255, n. 21).

shows results for simple bivariate regressions of the proximity variable on the covariates used in the main analysis. Some of these regressions yield statistically significant coefficients — for example, a case heard by a judge at the very end of their term is 1.9 percentage points more likely to feature a defendant who pleads guilty, compared to a judge at minimal distance from election. However, these predictive relationships are small in magnitude. When regressing the proximity measure on all of the covariates, the F-test exceeds the standard threshold of statistical significance given the size of the sample. However, the R-squared of the multivariate regression in Column 11 is minuscule (0.006). We take these tests of covariate balance to mean that deviations from “random” assignment are greater than what would occur by chance, but these deviations are so mild that the design can plausibly be characterized as a quasi-experiment.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Guidelines Upper	0.000*** (0.000)										0.000 (0.000)
Guidelines Lower		0.000*** (0.000)									0.000 (0.000)
Male			0.000 (0.004)								-0.004 (0.004)
Non-White				-0.005* (0.003)							-0.005 (0.003)
Age					0.000 (0.000)						-0.001* (0.001)
Squared Age											0.000* (0.000)
Plea						0.018*** (0.003)					0.012*** (0.004)
Weapon Used							0.020*** (0.005)				0.011 (0.009)
Weapon Enhancement								0.018*** (0.004)			0.004 (0.009)
6th Ed.											0.031*** (0.003)
7th Ed.											0.044*** (0.004)
No. of Charges									0.001** (0.000)		0.000* (0.000)
Rape										0.016** (0.006)	0.006 (0.007)
Robbery										0.001 (0.003)	0.000 (0.003)
Num.Obs.	47 297	47 297	47 297	47 297	47 297	47 297	47 297	47 297	47 297	47 297	47 297
R2	0.001	0.001	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000	0.006
R2 Adj.	0.001	0.001	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000	0.005
RMSE	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28

Table 1: Correlation Between Proximity and Covariates

The offenses considered in the analysis in Table 4 are the same as [Huber and Gordon \(2004\)](#): robbery, rape, and aggravated assault. These offenses are considered serious enough to be potentially politically harmful if a judge were seen giving out an insufficiently harsh punishment. At the same time, these categories also allow a judge more discretion in sentencing than offenses like homicide. In cases where multiple counts were considered against the same defendant, only the most serious case is analyzed (as determined by the highest suggested minimum sentence). Cases heard by judges appointed to fill midterm vacancies are retained in our data, as these judges still experience the pressure of an upcoming election. Each column in Table 4 employs a battery of controls also drawn from [Huber and Gordon \(2004\)](#): the upper and lower end of the sentencing guidelines in the case; the sex, race, age, and squared age of the defendant; whether the defendant pleaded guilty⁵; the offense type; whether a weapon was used and whether there was a weapon enhancement; the edition of the sentencing guidelines applied to the case; and the number of charges in the case.

[Huber and Gordon \(2004\)](#) include year fixed effects to account for over-time changes in sentencing that are unrelated to electoral pressures. Models 1 and 2 in Table 4 report results using these year fixed effects, to allow for comparison between our results and theirs. However, as the key explanatory variable is a measure of time until the election — and because we seek to analyze the outcomes in cases across years in a judge’s term, not just within them — we drop the year fixed effect in the remaining models to maximize treatment variation and therefore power. Columns 5 and 6 compromise by including a linear control that indicates where in our twenty years of out-of-sample replication data the case falls. As observed by [Huber and Gordon \(2004, p. 253\)](#), controlling for judge characteristics seems unnecessary, as they would not be expected to be correlated with electoral proximity. However, the inclusion of judge fixed effects does seem to be consequential, as Table 2 in [Huber and Gordon \(2004\)](#) shows they reduce the effect size of electoral proximity by approximately one-third. We include

⁵[Huber and Gordon \(2004\)](#) consider negotiated guilty pleas and non-negotiated guilty pleas separately. We do not make this distinction, as there are few non-negotiated pleas but a considerable number of *nolo contendere* pleas, which are not clearly labeled as negotiated or non-negotiated.

	Our Analysis	Huber & Gordon
Years	2000-2021	1990-1999
N	47,297	22,095
Mean Sentence (Months)	30.4	24.8
SD of sentences	35.64	28.73
No. of judges	426	265

Table 2: Description of Data Used in Main Analysis

models both with and without these fixed effects to facilitate comparison between [Huber and Gordon \(2004\)](#)’s analysis and our own. Robust standard errors are clustered by judge to reflect the correlated assignment of the electoral proximity treatment to each judge’s cases.⁶

In addition to replicating the model from [Huber and Gordon \(2004\)](#), we extend the analysis to an array of other model specifications drawn from and building upon other studies of electoral cycles in sentencing. Specifically, we adapt models from [Berdejó and Yuchtman \(2013\)](#)’s study of sentencing cycles in Washington (which were extended to more states in [Dippel and Poyker \(2021\)](#)). The models in Table 5 include the case-level, judge-level, and county-level controls specified in [Berdejó and Yuchtman \(2013\)](#): the race, gender, age, and prior criminal record of the defendant; whether the case was resolved in a plea deal; the level of unemployment; the crime the defendant was charged with; and judge fixed effects. Rather than control for the suggested upper and lower sentence, the models use a variable indicating each case’s sentencing guidelines cell. In Pennsylvania, the cell in the sentencing guidelines is defined by the offender’s prior record and the offense’s seriousness level. Cases in unusual cells (fewer than 150 observations) were grouped into dummy categories, constructed by sorting cases by the suggested minimum sentence from their cell, then aggregating them into bins of 150 observations each. We drop the year and quarter variables used in [Berdejó and Yuchtman \(2013\)](#) for the same reasons we drop the year fixed effect from the [Huber and Gordon \(2004\)](#) models. In addition, controls for the level of crime are

⁶As electoral cycles operate at the level of a judge-term combination, we also report a version of our results with judge-term clustering in Table A.2 in the Appendix.

included in the [Berdejó and Yuchtman \(2013\)](#) models but excluded in all but one of our models. Concerns about the accuracy of UCR reports abound (see review in [Loftin and McDowall \(2010\)](#)), and reports from Pennsylvania are not available for all years in our sample.

Model 1 is a baseline model using the variables described in the paragraphs above with the selection of offenses from [Huber and Gordon \(2004\)](#) (aggravated assault, rape, and robbery). Model 2 includes all FBI index violent crimes, the case selection from [Berdejó and Yuchtman \(2013\)](#). [Berdejó and Yuchtman \(2013\)](#) also assess the robustness of results when only considering plea deals, “where one might expect judicial discretion to be more limited” (p.748). The vast majority of cases involve guilty pleas (91% in our data). We include Model 3, which only includes cases that end in a plea deal, with the acknowledgment that plea deals are post-treatment and raise endogeneity concerns if defense attorneys are more likely to pursue plea deals with judges at certain points in their term. As sentence length is highly right-skewed, Model 4 uses a log of the sentence as an outcome variable. In case the process of grouping observations in unusual sentencing grid cells skews the analysis, Model 5 replaces the factor variable for sentencing guidelines cell with controls for the lower and upper end of the suggested sentence according to the sentencing guidelines. Model 6 in Table 5 includes controls for the annual crime rate and violent crime rate, which requires limiting the sample to the years 2001 through 2018.

The final model in this additional analysis stems from concerns about the restriction of both [Huber and Gordon \(2004\)](#)’s and [Berdejó and Yuchtman \(2013\)](#)’s analysis to only a few specific offense categories. While the offenses studied in previous work on cycle effects are undeniably serious, even the FBI’s list of index violent crimes excludes offenses which may be damaging to a judge who is deemed too lenient — including crimes against children, arson, drug trafficking, and firearms offenses. To more comprehensively examine serious crimes, we widen the list of offenses for Model 7 in Table 5 to all cases with an offense seriousness grade of 7 or greater — 7 being the

Statistic	Huber & Gordon Selection	Severity Selection
N	47,297	127,718
Mean Sentence (Months)	30.40	37.19
Median Sentence (Months)	18	12
St. Dev. of Sentences	35.64	88.65
No. of Judges	426	447

Huber & Gordon Selection includes cases of aggravated assault, rape, and robbery. Severity selection includes cases where the felony grade is 7 or higher. 7 is the median grade for aggravated assault, the lowest median grade of the three offenses included in the Huber & Gordon analysis.

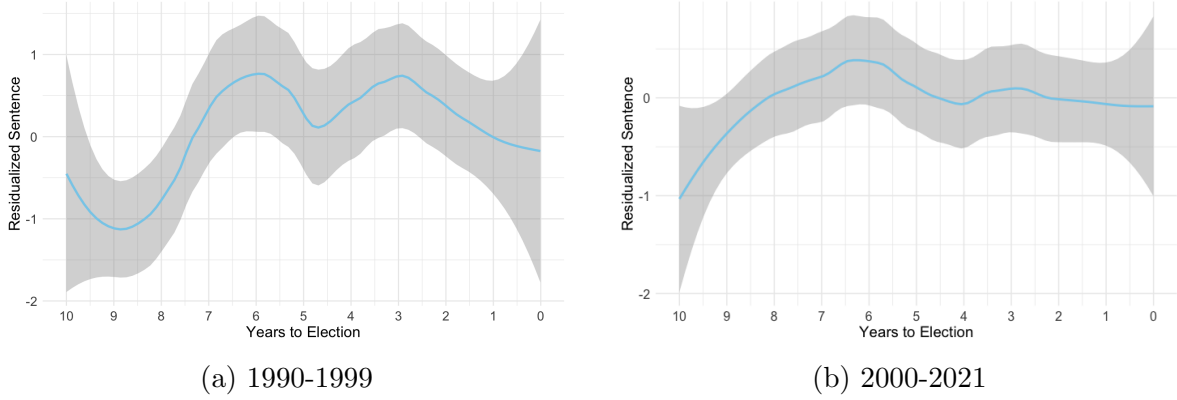
Table 3: Summary Statistics by Sample Selection Criteria

median seriousness for cases of aggravated assault, the lowest median seriousness level of the offenses included in Huber and Gordon’s analysis. A description of the expanded dataset is found in Table 3.

Results

Do judges facing retention elections sentence more harshly as their election approaches? Figure 1b depicts the relative harshness of judges in Pennsylvania over the course of their terms. The vertical axis of this plot represents the variation in sentencing that is not explained by relevant covariates. The horizontal axis indicates the years until the judge’s next election. If sentences exhibited cycles related to electoral proximity, we would expect to see a positive relationship, with residual variation in sentencing rising over time as the election draws closer. Specifically, the effect should emerge shortly before election day, when voters are most likely to be evaluating judicial performance and have had less time to forget potentially damaging information (Huber and Gordon 2004, p. 249). However, Figure 1b reveals little in the way of systematic variation between sentencing and electoral proximity. If anything, we see a dip in punitiveness at the beginning of the term, rather than an increase in punitiveness at the end of the term.

The results of the statistical analysis of out-of-sample data are reported in Table



LOESS line with 95% confidence intervals fitted to the residualized value of each sentence, controlling for the covariates from the [Huber and Gordon \(2004\)](#) model with judge fixed effects and a control for linear time (Model 6 in Table 4). This model includes aggravated assault, rape, and robbery cases only. Left figure includes data from 1990-1999, the years studied in [Huber and Gordon \(2004\)](#)’s original article. Figure on the right includes data from new, out-of-sample data from 2000-2021. The horizontal axis is elapsed time into the judge’s term (Proximity).

Figure 1: The Effects of Election Proximity on Judges’ Sentencing Severity

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	−0.576 (0.570)	−0.064 (0.501)	−0.439 (0.564)	0.132 (0.513)	−0.456 (0.560)	0.039 (0.506)
Num.Obs.	47 297	47 297	47 297	47 297	47 297	47 297
R2	0.689	0.714	0.688	0.713	0.688	0.713
R2 Adj.	0.689	0.711	0.688	0.710	0.688	0.711
RMSE	19.88	19.06	19.89	19.08	19.89	19.08
Time Control	Year FE	Year FE	No	No	Linear	Linear
Judge FE	No	Yes	No	Yes	No	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses are clustered at the judge level. Proximity is a linear measure of time until the next retention election. A proximity of 0 is equivalent to the first day of a judge’s term, while a proximity of 1 is the last day of the term, ten years later. A positive coefficient would be consistent with the story of election cycles in sentencing. “Linear” time control denotes the temporal position of each observation within the dataset, capturing how far into the 20-year span of 2001-2021 the observation occurs. Covariates included in each model are: upper and lower end of the sentencing guidelines in the case; age, squared age, sex, and race of the defendant; number of charges in the case; whether the defendant pled guilty; if the case included a weapon enhancement and whether a deadly weapon was used; the sentencing guidelines in place at the time of the case; and the type of offense. Table [B.1](#) is an expanded table that reports coefficients for all covariates.

Table 4: Results Replication Using Huber & Gordon Models

4. The explanatory variable *Proximity* is coded from 0 to 1, with 0 representing the greatest amount of time until election day, and 1 representing the least. The coefficient on *Proximity* can be interpreted as the change in sentence length as a case moves from the lowest level of electoral proximity to the highest. If there is a cycle effect in sentencing, the coefficient would be positive. Columns 1 and 2 replicate the model specifications from columns 1 and 3 in Table 2 of [Huber and Gordon \(2004\)](#). Columns 3-6 make slight changes to judge and year controls in each model. None of the coefficients on the *Proximity* regressor in any of the model specifications reach conventional levels of statistical significance. We find no clear pattern in the direction of the coefficients, with four positive coefficients (the expected direction) and three negative coefficients. The regression estimates offer little support for the hypothesis that judges sentence more harshly as their retention election gets closer. Table 5 shows regression results of seven additional model specifications, fit to models adapted from [Berdejó and Yuchtman \(2013\)](#). To account for the possibility of a non-linear relationship between electoral proximity and sentence length, Table A.9 in the Appendix presents results from a series of models adapted from the model specifications in Table 4 with a Box-Cox transformation applied to the dependent variable.⁷ Again, none of the coefficients on the proximity measure reach traditional levels of statistical significance in the expected direction.

⁷The Box-Cox transformation addresses potential non-linearities and non-normal error distributions by defining a flexible functional form for the dependent variable, $Y^{(\lambda)} = \frac{Y^\lambda - 1}{\lambda}$ when $\lambda \neq 0$, and $Y^{(\lambda)} = \ln(Y)$ when $\lambda = 0$. Because minimum sentence lengths include values of zero months and the transformation strictly requires $Y > 0$, we apply a standard scaling shift of +0.01 months to the outcome variable prior to estimation. For each specification, the parameter λ is estimated independently via maximum likelihood estimation over a search grid of $[-2, 2]$. The optimal λ values chosen to maximize normality and homoscedasticity are reported at the bottom of Table A.9.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Proximity	0.052 (0.411)	-0.171 (0.500)	-0.018 (0.371)	0.021 (0.341)	-0.058 (0.036)	0.034 (0.436)	0.062 (0.276)
Num.Obs.	47 297	52 002	38 942	47 297	47 297	43 966	125 110
R2	0.550	0.947	0.575	0.716	0.345	0.542	0.917
R2 Adj.	0.544	0.947	0.569	0.713	0.337	0.537	0.917
RMSE	23.91	29.18	18.94	18.98	2.07	24.08	25.49
Case Selection	H&G	B&Y	H&G	H&G	H&G	H&G	Severity
Adjudication	All	All	Pleas	All	All	All	All
Outcome	Topcoded	Topcoded	Topcoded	Log	Topcoded	Topcoded	Topcoded
Guidelines	Cell	Cell	Cell	Cell	Upper/Lower	Cell	Cell
Crime Rates	N	N	N	N	N	Y	N
County FE	Y	Y	Y	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y	Y	Y	Y

Robust standard errors in parentheses are clustered at the judge level. Proximity is a linear measure of time until the next retention election. A proximity of 0 is equivalent to the first day of a judge's term, while a proximity of 1 is the last day of the term, ten years later. For case selection, "H&G" includes offenses from the [Huber and Gordon \(2004\)](#) article (aggravated assault, robbery, and rape), while "B&Y" includes additional offenses specified by [Berdejó and Yuchtman \(2013\)](#) (murder, manslaughter). "Severity" includes all class 1 and 2 felonies along with murder and manslaughter. For adjudication, "all" refers to cases which went to trial or resulted in a guilty plea, while "pleas" excludes trials. For outcome, "topcoded" censors the sentence at 720 months, while "log" is a log of the topcoded outcome. For guidelines, "cell" indicates a factor control variable for the cell of the sentencing guidelines, while "upper/lower" includes separate numeric controls for the upper and lower suggested sentence for the cell. Crime rates indicates the presence of controls for county level violent and total crime rates.

Table 5: Additional Model Specifications

Taken together, our results suggest an effect size close to zero. Table 6 formally compares these results to those in the period studied by [Huber and Gordon \(2004\)](#). For five of our six model specifications, the point estimates for (1) the effect of electoral pressure on sentencing during the 2000-2021 period and (2) for the effect of electoral pressure during the 1990-1999 period are statistically distinguishable from each other. To summarize, the out-of-sample data generate coefficients that hover around zero, and even the largest coefficient (0.132) is just a fraction of its counterpart from the original period (2.508).

Discussion

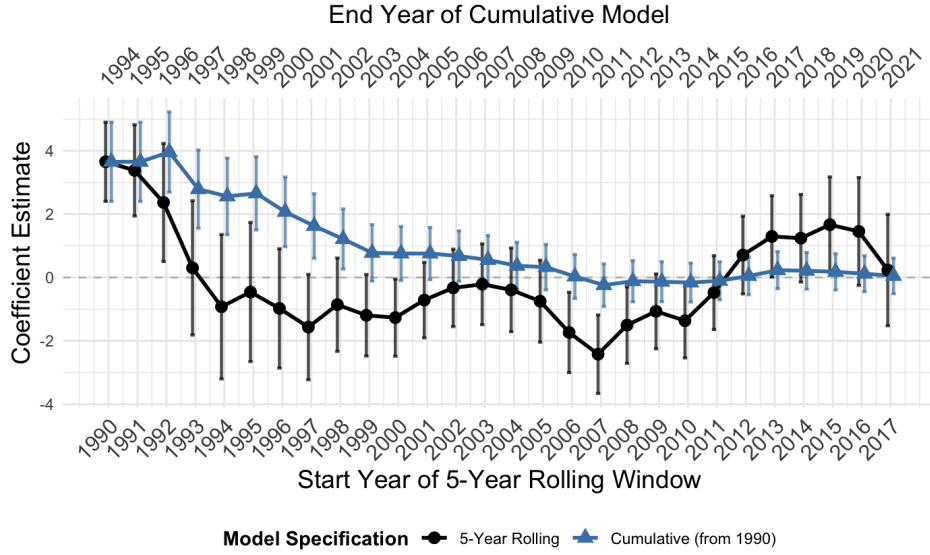
Our analysis of newly available data from Pennsylvania offers little evidence of election cycles in sentencing across a variety of model specifications. The results from the 2001-2021 dataset are also significantly different from the sizable effects we find using data from 1990-1999 (see Table 6). How then do we interpret the differences between our findings and those of [Huber and Gordon \(2004\)](#)? Having found similar results in our re-analysis of the 1990-1999 data (Tables A.5 and A.6 in the Appendix), we exclude the possibility of researcher error. Next, we consider whether the effects found in the [Huber and Gordon \(2004\)](#) analysis were specific to the era they studied. Figure 2 plots over-time estimates for the coefficient on *Proximity* from two different linear model specifications. One model is fit to a rolling five-year subset of the data, while the other is fit to a cumulative subset of the data. Both employ the battery of covariates from [Huber and Gordon \(2004\)](#). Running the model on successive five-year subsets of the data, we find that the strong positive connection between electoral proximity and sentence length found in [Huber and Gordon \(2004\)](#) is driven entirely by the data from the early 1990s. When estimating the effect from a cumulative subset of the data, the effect size begins significant and positive, but is reduced to zero as more data from

Model	Coefficient _{2000–2021}	Coefficient _{1990–1999}	Difference in Coefs.
1	-0.576 (0.570)	2.164 (1.065)	-2.74** (1.208)
2	0.064 (0.501)	1.688 (0.947)	-1.624 (1.072)
3	-0.439 (0.564)	2.693 (1.028)	-3.132*** (1.173)
4	0.132 (0.513)	2.508 (0.938)	-2.376** (1.069)
5	-0.456 (0.560)	2.599 (1.017)	-3.055*** (1.161)
6	0.039 (0.506)	2.020 (0.959)	-1.981* (1.085)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Comparison of linear regression results based on sentencing data from 1990–1999 and 2000–2021. Model 1 includes year fixed effects, Model 2 includes year and judge fixed effects. Model 3 includes judge fixed effects. Model 4 includes no fixed effects. Model 5 includes a control for linear time. Model 6 includes a control for linear time and judge fixed effects. Coefficients and standard errors for the 1990-1999 period taken from the authors’ re-analysis of the data in Table A.5 in the Appendix. Values for the 2000-2021 data come from Table 4.

Table 6: Comparison of Out-of-Sample Replication Estimates from 2000-2021 to Regression Estimates based on the 1990-1999 Period



Each point in black represents the linear estimate for “Proximity” from a model run on successive five-year subsets of data, with 95% confidence intervals. The lower x-axis represents the start year of the five-year rolling window. Each point in blue represents the linear estimate for “Proximity” from a model run on a cumulative subset of the data beginning in 1990, also with 95% confidence intervals. The upper x-axis represents the end year of the data used in the cumulative model. The y-axis is the value of the estimate. Positive estimates indicate longer sentences as election day gets closer. Estimates drawn from a linear model with a fixed effect for sentencing year, but without judge fixed effects to maximize variation in the explanatory variable (Model 1 in Table 4). In periods covering only one set of sentencing guidelines, the covariate for the guidelines is dropped.

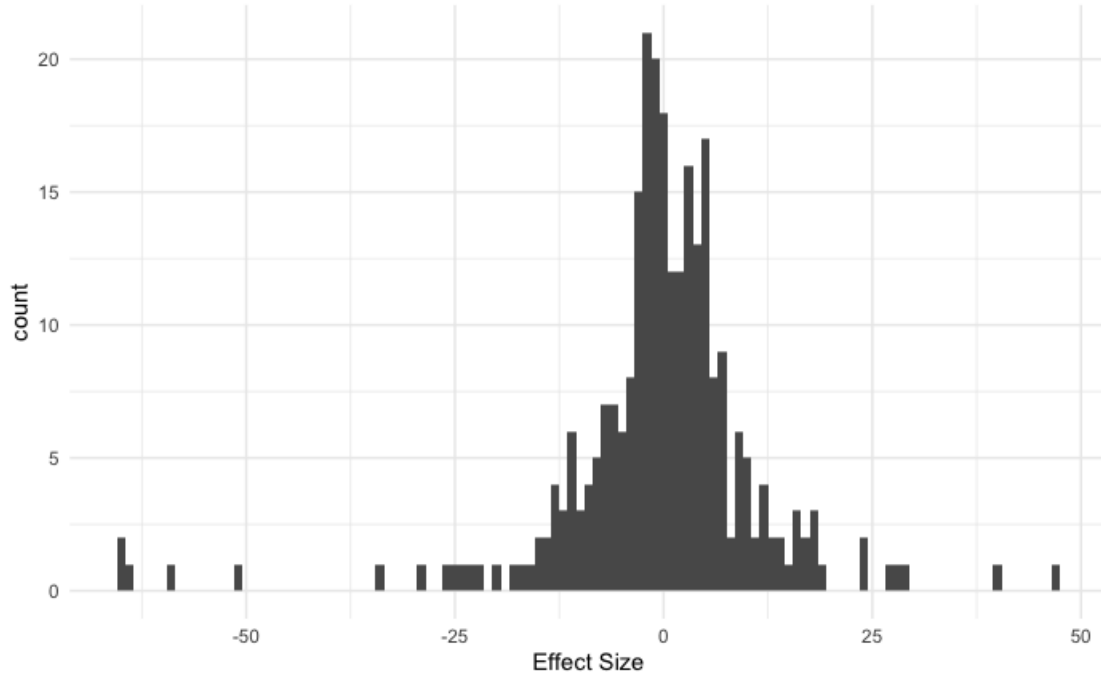
Figure 2: Rolling Linear Estimate of the Effect of Proximity on Sentence Length

the 2000s is included.⁸ In sum, the results indicate that the average effect of electoral proximity on sentence lengths is statistically indistinguishable from zero, although the cycle effect may have driven more punitive sentences in the early 1990s.

Conclusion

In an attempt to take a fresh look at electoral influences on the courts, we set out to replicate a major finding of judicial cycle effects using out-of-sample data. Employing over two decades of new data from Pennsylvania criminal courts, we find little support for the claim that electoral proximity increases sentence length in serious cases. The meaning of these null results remains open to interpretation. [Huber and Gordon](#)

⁸Table A.8 in the Appendix reports null estimates for models run on the pooled dataset spanning 1990-2022



Horizontal axis is the size of the coefficient on “Proximity” in the OLS model of effects with a control for linear time (Column 5 in Table 4). The distribution of judge-specific effects is estimated based on the 274 judges who presided over 50 or more cases. The total number of cases meeting this criterion is 44,750, or 93.4% of the total pool of cases.

Figure 3: Estimated Effect Size for Models Run Separately on Each Judge

(2004) suggest that a null finding could indicate three different patterns of judicial behavior: judges being attentive to voters’ preference for punishment at all times, judges never being attentive to voters’ preferences, or “bidirectional convergence, wherein some judges become more conservative as election nears while others become more liberal, leading to zero net effect” (p. 255). Estimating the regression model with a linear time control (Model 5 from Table 4) separately on each judge, we observe a unimodal distribution centered on zero (Figure 3). While some outlier judges exhibit extreme swings towards longer or shorter sentencing towards the end of their terms, the middle 50% of effect sizes are between -4.04 and 4.75. However, a joint Wald test indicates that the dispersion of effect sizes is larger than we would expect due to estimation error alone ($p < 0.0001$). While sentencing behavior appears to genuinely vary with electoral proximity for some judges, the shape of the distribution calls into question the notion that distinct groups of liberal and conservative judges cancel each other out.

Further, these data cannot speak to why there is a significant cycle effect in the early 1990s but not in later years. Three distinct scenarios could yield these results. First, our null results may be a statistical anomaly, reliant on the specific years and state included in the sample. Second, [Huber and Gordon \(2004\)](#)’s significant results may be an anomaly for the same reason. Finally, both studies may identify genuine effects, and the difference between our results could reflect a genuine change in the politics of sentencing in Pennsylvania that occurred between the early 1990s and the following decades. To adjudicate between these explanations, more data will be needed. If our results are a statistical fluke, then there should be evidence of cycle effects in other retention states in recent years. If [Huber and Gordon \(2004\)](#) was a fluke, then there should be no evidence of cycle effects in other states in the 1900-1999 period. If there is a genuine period effect, cycle effects should be evident in many states in the 1990s, but absent in later years. At this time we can offer only one small piece of supporting evidence for our null effects — a preliminary analysis of sentencing trends in Colorado during the 2000-2024 period also yields no evidence of cycle effects (Table [A.10](#)). However, truly validating these null findings will require a greater body of evidence. Extending the analysis to the nine other states which hold retention elections in at least some jurisdictions will be critical to determine the true influence of electoral proximity on sentencing.⁹

⁹Only one other study ([Dippel and Poyker 2021](#)) explores the cycle effect across multiple retention states, though their replication sample contains unresolved data anomalies. In their analysis of Pennsylvania, [Dippel and Poyker \(2021\)](#) report a volume of annual observations that is disproportionately higher than both [Huber and Gordon \(2004\)](#) and our own sample. Furthermore, their estimated proximity effect of 15.214 months is over four times larger than the original effect size identified by [Huber and Gordon \(2004, p. 255\)](#) (≈ 3 to 4.25 months) and our own estimate of 3.522 months for the 1990–1994 window. Finally, this outsized estimate only achieves conventional statistical significance when utilizing robust standard errors rather than naive variations. [Dippel and Poyker \(2021\)](#) also analyze data from Colorado and find no evidence of cycle effects. Our analysis yields similar results, though we employ several additional years of data before and after the data used in their study (Table [A.10](#)).

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Appendices

A Additional Analyses

Descriptive Statistics by Year.....	p. A-2
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B Tables With All Covariates Reported

Linear Regression Results (Table 4).....	p. B-1
Additional Specifications (Table 5)	p. B-2

Statistic	1	2	3	4	5	6	7	8	9	10
Defendant Age (mean)	30.27	30.13	30.32	30.04	30.47	30.01	30.15	30.27	30.34	30.40
Minimum Sentence (mean)	29.11	29.64	30.29	30.35	29.99	30.08	30.54	30.92	31.63	31.44
Guidelines Lower (mean)	29.60	28.99	30.23	30.18	30.33	30.14	30.93	31.05	32.02	31.90
Guidelines Upper (mean)	41.84	40.79	42.35	42.52	42.56	42.50	43.58	43.92	45.13	44.73
# of Charges (mean)	2.68	2.62	2.62	2.81	2.71	2.59	2.69	2.86	3.00	2.77
% Male	88.90	88.40	89.30	88.60	88.40	89.00	88.40	88.40	88.20	88.90
% Plea	81.20	80.20	81.00	83.50	83.00	83.00	83.80	82.00	82.90	84.30
% Non-white	54.80	54.20	54.10	56.10	52.10	54.10	53.00	53.70	52.50	53.80
% Weapon Used	7.70	8.30	8.20	8.30	8.80	9.20	9.10	9.70	9.40	9.10
% Weapon Enhancement	10.60	10.40	10.50	10.40	10.90	11.90	11.30	11.90	11.60	12.20

Average values of covariates calculated from sentencing data from 2000-2021 with data divided by 0.1 increments in the Proximity measure.

Table A.1: Descriptive Statistics by Year of the Term

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	−0.576 (0.750)	−0.064 (0.490)	−0.439 (0.739)	0.132 (0.493)	−0.456 (0.740)	0.039 (0.491)
Guidelines Upper	0.133*** (0.019)	0.136*** (0.018)	0.132*** (0.019)	0.134*** (0.018)	0.131*** (0.019)	0.133*** (0.018)
Guidelines Lower	0.721*** (0.022)	0.731*** (0.021)	0.723*** (0.022)	0.734*** (0.021)	0.724*** (0.022)	0.734*** (0.021)
Male	4.592*** (0.300)	4.132*** (0.281)	4.611*** (0.304)	4.144*** (0.282)	4.593*** (0.302)	4.132*** (0.281)
Non-White	−0.446 (0.293)	1.118*** (0.208)	−0.405 (0.301)	1.115*** (0.208)	−0.419 (0.298)	1.108*** (0.207)
Age	0.093 (0.061)	0.078 (0.058)	0.093 (0.060)	0.075 (0.058)	0.089 (0.060)	0.072 (0.058)
Squared Age	−0.002** (0.001)	−0.002* (0.001)	−0.002** (0.001)	−0.002* (0.001)	−0.002** (0.001)	−0.002* (0.001)
Plea	−7.013*** (0.615)	−7.674*** (0.524)	−6.698*** (0.645)	−7.231*** (0.551)	−6.804*** (0.634)	−7.338*** (0.549)
Weapon Used	−0.298 (0.726)	0.275 (0.702)	−0.262 (0.728)	0.326 (0.707)	−0.304 (0.731)	0.284 (0.708)
Weapon Enhancement	1.038 (0.760)	−0.365 (0.629)	1.001 (0.768)	−0.389 (0.641)	1.042 (0.767)	−0.364 (0.640)
6th Ed.	−2.391** (0.828)	−2.947*** (0.776)	0.217 (0.535)	0.032 (0.456)	−0.937 (0.707)	−1.114+ (0.627)
7th Ed.	−3.596** (1.170)	−3.880*** (1.139)	−0.408 (0.647)	−0.763 (0.475)	−2.806** (1.000)	−3.136** (0.974)
No. of Charges	0.162+ (0.090)	0.174+ (0.100)	0.165+ (0.091)	0.178+ (0.102)	0.163+ (0.090)	0.176+ (0.101)
Rape	15.022*** (1.151)	13.088*** (1.098)	15.362*** (1.156)	13.474*** (1.107)	15.167*** (1.133)	13.272*** (1.094)
Robbery	3.453*** (0.264)	3.645*** (0.235)	3.430*** (0.261)	3.634*** (0.232)	3.447*** (0.261)	3.647*** (0.232)
Num.Obs.	47 297	47 297	47 297	47 297	47 297	47 297
R2	0.689	0.714	0.688	0.713	0.688	0.713
R2 Adj.	0.689	0.711	0.688	0.710	0.688	0.711
RMSE	19.88	19.06	19.89	19.08	19.89	19.08

Table A.2: Regression Models with Errors Clustered at Judge-Term Level

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	-0.552 (0.333)	-0.234 (0.459)	-0.414 (0.328)	0.098 (0.453)	-0.431 (0.328)	0.042 (0.453)
Guidelines Upper	0.133*** (0.015)	0.062** (0.020)	0.132*** (0.016)	0.061** (0.020)	0.131*** (0.016)	0.060** (0.020)
Guidelines Lower	0.722*** (0.019)	0.772*** (0.025)	0.724*** (0.019)	0.775*** (0.025)	0.724*** (0.019)	0.775*** (0.025)
Male	4.572*** (0.230)	4.454*** (0.296)	4.592*** (0.230)	4.479*** (0.296)	4.574*** (0.230)	4.450*** (0.296)
Non-White	-0.445* (0.187)	1.474*** (0.253)	-0.404* (0.187)	1.452*** (0.254)	-0.418* (0.187)	1.446*** (0.254)
Age	0.093 (0.056)	0.076 (0.067)	0.093 (0.056)	0.074 (0.068)	0.089 (0.056)	0.072 (0.068)
Squared Age	-0.002** (0.001)	-0.002* (0.001)	-0.002** (0.001)	-0.002* (0.001)	-0.002** (0.001)	-0.002* (0.001)
Plea	-7.026*** (0.313)	-7.985*** (0.416)	-6.704*** (0.307)	-7.273*** (0.409)	-6.809*** (0.310)	-7.423*** (0.411)
Weapon Used	-0.297 (0.654)	1.179 (0.876)	-0.261 (0.654)	1.268 (0.878)	-0.303 (0.655)	1.202 (0.879)
Weapon Enhancement	0.994 (0.574)	-0.294 (0.776)	0.956 (0.573)	-0.386 (0.777)	0.997 (0.574)	-0.345 (0.778)
6th Ed.	-2.416*** (0.668)	-3.101*** (0.827)	0.089 (0.222)	-0.339 (0.338)	-1.063** (0.370)	-1.596** (0.490)
7th Ed.	-3.636*** (0.933)	-3.180** (1.117)	-0.523* (0.241)	-0.851* (0.417)	-2.917*** (0.673)	-3.450*** (0.841)
No. of Charges	0.161 (0.088)	0.906*** (0.071)	0.163 (0.090)	0.919*** (0.071)	0.162 (0.089)	0.912*** (0.071)
Rape	14.973*** (0.794)	15.631*** (1.100)	15.301*** (0.799)	15.917*** (1.102)	15.107*** (0.796)	15.697*** (1.101)
Robbery	3.487*** (0.181)	3.613*** (0.237)	3.462*** (0.181)	3.583*** (0.238)	3.480*** (0.181)	3.594*** (0.238)
Num.Obs.	47 297	29 117	47 297	29 117	47 297	29 117
RMSE	19.88	20.15	19.89	20.19	19.89	20.18
Time Control	Year FE	Year FE	No	No	Linear	Linear
Judge Effects	No	Yes	No	Yes	No	Yes
Judges	All	Top	All	Top	All	Top

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Two limit tobit models adapted from [Huber and Gordon \(2004\)](#) with the outcome left-censored at 0 months (or the mandatory minimum sentence) and right-censored at half the maximum sentence (p. 251). Judge effects models substitute each covariate with the residual of the covariate regressed on judge fixed effects. Covariates included in each model are: upper and lower end of the sentencing guidelines in the case; age, squared age, sex, and race of the defendant; number of charges in the case; whether the defendant pled guilty; if the case included a weapon enhancement and whether a deadly weapon was used; the sentencing guidelines in place at the time of the case; and the type of offense. Judge effects models includes cases heard by judges in the top quartile in terms of cases heard (84 judges, with a median of 239.5 cases each).

Table A.3: Two-Limit Tobit Models

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	−0.068 (0.044)	−0.025 (0.036)	−0.055 (0.043)	−0.008 (0.036)	−0.056 (0.043)	−0.014 (0.036)
Def. Male	0.301*** (0.024)	0.271*** (0.021)	0.301*** (0.024)	0.272*** (0.021)	0.301*** (0.024)	0.271*** (0.021)
Def. Non-White	−0.075** (0.025)	0.064*** (0.017)	−0.073** (0.026)	0.065*** (0.017)	−0.074** (0.026)	0.064*** (0.017)
Def. Age	−0.011 (0.005)	−0.011** (0.004)	−0.011* (0.005)	−0.011** (0.004)	−0.011* (0.005)	−0.011** (0.004)
Def. Age Sq.	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Plea	−0.393*** (0.047)	−0.448*** (0.039)	−0.382*** (0.046)	−0.426*** (0.040)	−0.387*** (0.046)	−0.432*** (0.040)
Weapon Used	−0.163** (0.052)	−0.101* (0.050)	−0.161** (0.052)	−0.097 (0.050)	−0.163** (0.053)	−0.100* (0.051)
Weapon En.	−0.126* (0.054)	−0.222*** (0.045)	−0.123* (0.054)	−0.219*** (0.045)	−0.122* (0.054)	−0.219*** (0.045)
No. of Counts	0.006 (0.004)	0.007 (0.005)	0.006 (0.004)	0.007 (0.005)	0.006 (0.004)	0.007 (0.005)
6th Ed.	−0.169** (0.064)	−0.221*** (0.060)	0.010 (0.041)	−0.004 (0.035)	−0.041 (0.046)	−0.069 (0.041)
7th Ed.	−0.201* (0.079)	−0.247*** (0.073)	−0.032 (0.053)	−0.057 (0.040)	−0.140* (0.070)	−0.193** (0.064)
Rape	0.608*** (0.057)	0.504*** (0.062)	0.627*** (0.059)	0.527*** (0.063)	0.615*** (0.057)	0.511*** (0.063)
Robbery	0.284*** (0.020)	0.301*** (0.017)	0.283*** (0.020)	0.299*** (0.017)	0.284*** (0.020)	0.300*** (0.017)
Num.Obs.	47 798	47 798	47 798	47 798	47 798	47 798
R2	0.041	0.128	0.039	0.126	0.039	0.127
R2 Adj.	0.040	0.120	0.039	0.118	0.039	0.119
RMSE	1.48	1.41	1.48	1.41	1.48	1.41
Time Control	Year FE	Year FE	No	No	Linear	Linear
Judge FE	No	Yes	No	Yes	No	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Same models as reported in Table 4, with standardized sentence. The standardized sentence variable is the difference between the minimum sentence and the lower end of the guidelines over the difference between the upper end of the guidelines and the lower end of the guidelines. Where the upper end of the guidelines and the lower end of the guidelines are the same, and the minimum sentence matches the guidelines, the standardized sentence is zero. Robust standard errors are reported in parentheses and are clustered at the judge level.

Table A.4: Main Model Results with Standardized Dependent Variable

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	2.072 (1.080)	1.522 (0.989)	2.661* (1.034)	2.439* (0.972)	2.601* (1.030)	1.955 (1.002)
Gl. Lower	0.459*** (0.079)	0.474*** (0.075)	0.447*** (0.080)	0.463*** (0.075)	0.451*** (0.079)	0.471*** (0.075)
Gl. Upper	0.329*** (0.063)	0.311*** (0.059)	0.344*** (0.065)	0.326*** (0.060)	0.338*** (0.064)	0.315*** (0.059)
Male	5.724*** (0.489)	5.309*** (0.506)	5.689*** (0.486)	5.282*** (0.504)	5.744*** (0.486)	5.312*** (0.504)
Non-White	1.571** (0.526)	2.653*** (0.451)	1.557** (0.542)	2.655*** (0.449)	1.647** (0.543)	2.653*** (0.457)
Age	0.023 (0.087)	-0.002 (0.089)	0.026 (0.087)	-0.005 (0.088)	0.016 (0.088)	-0.006 (0.089)
Age Sq.	-0.002 (0.001)	-0.001 (0.001)	-0.002 (0.001)	-0.001 (0.001)	-0.002 (0.001)	-0.001 (0.001)
Plea	-1.482* (0.604)	-0.335 (0.655)	-1.603** (0.612)	-0.368 (0.638)	-1.406* (0.600)	-0.332 (0.656)
Weapon En.	20.839*** (1.038)	20.557*** (0.977)	20.996*** (1.071)	20.686*** (0.990)	20.914*** (1.069)	20.603*** (0.997)
1994 Gl.	-3.382* (1.609)	-2.732 (1.618)	5.128*** (0.958)	3.989*** (0.896)	-0.353 (1.125)	-0.990 (1.092)
1997 Gl.	-11.391*** (2.057)	-9.871*** (1.815)	1.747** (0.604)	0.882 (0.699)	-6.212*** (1.428)	-6.575*** (1.328)
No. of Charges	0.556 (0.358)	0.539 (0.373)	0.561 (0.365)	0.543 (0.378)	0.558 (0.361)	0.539 (0.374)
Rape	15.165*** (1.107)	14.533*** (1.099)	15.504*** (1.124)	14.821*** (1.126)	15.118*** (1.116)	14.484*** (1.106)
Robbery	6.306*** (0.484)	6.587*** (0.499)	6.436*** (0.497)	6.683*** (0.506)	6.321*** (0.488)	6.595*** (0.501)
Num.Obs.	17 358	17 358	17 358	17 358	17 358	17 358
R2	0.548	0.575	0.544	0.572	0.547	0.574
R2 Adj.	0.548	0.566	0.544	0.564	0.546	0.566
RMSE	19.77	19.17	19.86	19.24	19.80	19.19
Time Control	Year FE	Year FE	No	No	Linear	Linear
Judge FE	No	Yes	No	Yes	No	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

* p < 0.1, ** p < 0.05, *** p < 0.01

Robust standard errors in parentheses are clustered at the judge level. Proximity is a linear measure of time until the next retention election. A proximity of 0 is equivalent to the first day of a judge's term, while a proximity of 1 is the last day of the term, ten years later. A positive coefficient would be consistent with the story of election cycles in sentencing. "Linear" time control denotes the temporal position of each observation within the dataset, capturing how far into the 10-year span of 1990-1999 the observation occurs. No data is available for 1995. Covariates included in each model are: upper and lower end of the sentencing guidelines in the case; age, squared age, sex, and race of the defendant; number of charges in the case; whether the defendant pled guilty; if the case included a weapon enhancement and whether a deadly weapon was used; the sentencing guidelines in place at the time of the case; and the type of offense.

Table A.5: Linear Regression Models Run On Data From Huber & Gordon (2004)

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	2.282*** (0.567)	1.725** (0.644)	2.841*** (0.557)	2.510*** (0.631)	2.796*** (0.556)	2.117*** (0.632)
Gl. Lower	0.448*** (0.043)	0.460*** (0.042)	0.434*** (0.043)	0.448*** (0.042)	0.438*** (0.043)	0.455*** (0.042)
Gl. Upper	0.341*** (0.034)	0.323*** (0.034)	0.357*** (0.034)	0.337*** (0.034)	0.350*** (0.034)	0.328*** (0.034)
Male	5.795*** (0.553)	5.416*** (0.542)	5.783*** (0.555)	5.398*** (0.543)	5.822*** (0.554)	5.420*** (0.542)
Non-White	1.581*** (0.331)	2.650*** (0.362)	1.592*** (0.332)	2.655*** (0.363)	1.663*** (0.331)	2.652*** (0.362)
Age	0.024 (0.077)	-0.004 (0.075)	0.025 (0.077)	-0.005 (0.075)	0.017 (0.077)	-0.007 (0.075)
Age Sq.	-0.002 (0.001)	-0.001 (0.001)	-0.002 (0.001)	-0.001 (0.001)	-0.002 (0.001)	-0.001 (0.001)
Plea	-1.540*** (0.358)	-0.367 (0.374)	-1.574*** (0.358)	-0.365 (0.375)	-1.422*** (0.358)	-0.337 (0.374)
Weapon En.	20.484*** (0.506)	20.276*** (0.504)	20.648*** (0.508)	20.408*** (0.504)	20.588*** (0.507)	20.342*** (0.504)
1994 Gl.	-3.195** (1.150)	-2.531* (1.135)	4.868*** (0.401)	3.852*** (0.460)	0.519 (0.686)	-0.266 (0.712)
1997 Gl.	-11.169*** (1.329)	-9.594*** (1.316)	1.437*** (0.412)	0.756 (0.475)	-4.884*** (0.908)	-5.412*** (0.943)
No. of Charges	0.548*** (0.038)	0.533*** (0.037)	0.552*** (0.038)	0.536*** (0.037)	0.550*** (0.038)	0.534*** (0.037)
Rape	14.955*** (0.703)	14.409*** (0.694)	15.229*** (0.704)	14.657*** (0.694)	14.937*** (0.703)	14.390*** (0.694)
Robbery	6.358*** (0.334)	6.658*** (0.332)	6.457*** (0.335)	6.735*** (0.333)	6.367*** (0.334)	6.662*** (0.332)
Num.Obs.	17 358	17 358	17 358	17 358	17 358	17 358
RMSE	19.78	19.21	19.86	19.27	19.81	19.22
Time Control	Year FE	Year FE	No	No	Linear	Linear
Judge FE	No	Yes	No	Yes	No	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Two limit tobit models adapted from [Huber and Gordon \(2004\)](#) with the outcome left-censored at 0 months (or the mandatory minimum sentence) and right-censored at half the maximum sentence (p. 251). “Linear” time control denotes the temporal position of each observation within the dataset, capturing how far into the 10-year span of 1990-1999 the observation occurs. No data is available for 1995. Covariates included in each model are: upper and lower end of the sentencing guidelines in the case; age, squared age, sex, and race of the defendant; number of charges in the case; whether the defendant pled guilty; if the case included a weapon enhancement and whether a deadly weapon was used; the sentencing guidelines in place at the time of the case; and the type of offense.

Table A.6: Tobit Models Run On Data From Huber & Gordon (2004)

	Last four years	First four years
Proximity	2.053 (1.661)	−0.915 (2.071)
Guidelines Upper	0.132*** (0.026)	0.160*** (0.034)
Guidelines Lower	0.739*** (0.030)	0.695*** (0.035)
Male	3.841*** (0.391)	3.958*** (0.441)
Non-White	1.255*** (0.336)	1.360*** (0.382)
Age	−0.068 (0.087)	0.252* (0.110)
Squared Age	0.001 (0.001)	−0.004** (0.002)
Plea	−6.977*** (0.649)	−8.441*** (0.821)
Weapon Used	−0.009 (1.096)	0.295 (1.025)
Weapon Enhancement	0.486 (0.848)	−0.679 (0.992)
6th Ed.	0.214 (0.604)	0.000 (0.807)
7th Ed.	0.260 (0.674)	−1.373 (0.976)
No. of Charges	0.628*** (0.093)	0.113 (0.070)
Rape	11.072*** (1.280)	13.833*** (1.882)
Robbery	3.643*** (0.329)	3.468*** (0.371)
Num.Obs.	19 599	17 526
R2	0.720	0.718
R2 Adj.	0.714	0.712
RMSE	18.77	19.11

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Regression specification is Model 4 in Table 4. Robust standard errors are reported in parentheses and are clustered at the judge level. Proximity is a linear measure of time until the next retention election. A proximity of 0 is equivalent to the first day of a judge’s term, while a proximity of 1 is the last day of the term, ten years later. “Last four years” (“First four years”) restricts the sample to cases heard in the last (first) four years of all judges’ terms.

Table A.7: Estimating Cycle Effects for Judges who Are Serving Early or Late in their Terms

One explanation for these null results may be the length of the judge’s terms. Pennsylvania Common Pleas judges serve ten-year terms, giving them longer to shirk at the beginning of the term. If electoral pressure does not appear to increase linearly by year, but instead ramps up exclusively in the last few years before election, then looking at the entire ten-year cycle may hide the effect. (Because term lengths were ten years in the period that Huber and Gordon study, this factor alone cannot explain our divergent findings.) We test this possibility by restricting the baseline model to Pennsylvania cases heard in the last four years of each judge’s term. We do the same with cases heard in the first four years of the term as a placebo test. The first two columns in Table [A.7](#) reveal little difference between the two sets of estimates. We find no evidence of a cycle effect in sentencing when restricting the cases to the years closest to re-election.

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	0.049 (0.524)	0.090 (0.459)	0.358 (0.521)	0.494 (0.469)	0.408 (0.521)	0.454 (0.466)
Gl. Lower	0.688*** (0.023)	0.700*** (0.022)	0.689*** (0.023)	0.702*** (0.022)	0.689*** (0.023)	0.702*** (0.022)
Gl. Upper	0.153*** (0.020)	0.151*** (0.019)	0.153*** (0.020)	0.150*** (0.019)	0.152*** (0.020)	0.149*** (0.019)
Male	4.833*** (0.278)	4.386*** (0.267)	4.849*** (0.282)	4.393*** (0.268)	4.836*** (0.280)	4.386*** (0.267)
Non-White	0.302 (0.304)	1.545*** (0.210)	0.328 (0.311)	1.543*** (0.210)	0.318 (0.310)	1.538*** (0.209)
Age	0.090 (0.056)	0.068 (0.052)	0.092 (0.056)	0.065 (0.052)	0.087 (0.056)	0.063 (0.052)
Age Sq.	-0.002** (0.001)	-0.002* (0.001)	-0.002** (0.001)	-0.002* (0.001)	-0.002** (0.001)	-0.002* (0.001)
Plea	-5.019*** (0.519)	-5.016*** (0.469)	-4.921*** (0.472)	-4.914*** (0.434)	-5.138*** (0.478)	-5.094*** (0.441)
Weapon En.	6.521*** (0.829)	6.033*** (0.774)	6.597*** (0.846)	6.104*** (0.783)	6.630*** (0.849)	6.115*** (0.786)
1994 Gl.	-2.057 (1.742)	-1.111 (1.699)	7.026*** (1.056)	5.975*** (0.952)	5.904*** (1.084)	5.029*** (0.959)
1997 Gl.	-12.194*** (2.499)	-10.190*** (1.971)	2.299*** (0.633)	1.336 (0.756)	0.018 (0.918)	-0.567 (0.913)
2005 Gl.	-14.794*** (2.628)	-13.183*** (2.130)	2.114** (0.650)	0.834 (0.905)	-1.768 (1.216)	-2.407* (1.217)
2012 Gl.	-15.996*** (2.769)	-14.170*** (2.302)	1.468* (0.669)	-0.102 (0.949)	-3.910* (1.574)	-4.607** (1.557)
No. of Charges	0.232* (0.115)	0.238* (0.122)	0.235* (0.116)	0.240* (0.123)	0.233* (0.115)	0.239* (0.122)
Rape	14.985*** (0.943)	13.520*** (0.908)	15.345*** (0.944)	13.853*** (0.913)	15.144*** (0.935)	13.678*** (0.906)
Robbery	4.141*** (0.277)	4.352*** (0.262)	4.141*** (0.281)	4.355*** (0.266)	4.147*** (0.280)	4.356*** (0.265)
Num.Obs.	64 655	64 655	64 655	64 655	64 655	64 655
R2	0.652	0.677	0.651	0.675	0.651	0.676
R2 Adj.	0.652	0.673	0.651	0.672	0.651	0.672
RMSE	20.12	19.41	20.16	19.44	20.16	19.44
Time Control	Year FE	Year FE	No	No	Linear	Linear
Judge FE	No	Yes	No	Yes	No	Yes
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors in parentheses are clustered at the judge level. Proximity is a linear measure of time until the next retention election. A proximity of 0 is equivalent to the first day of a judge's term, while a proximity of 1 is the last day of the term, ten years later. A positive coefficient would be consistent with the story of election cycles in sentencing. "Linear" time control denotes the temporal position of each observation within the dataset, capturing how far into the 31-year span of 1990-2021 the observation occurs. Covariates included in each model are: upper and lower end of the sentencing guidelines in the case; age, squared age, sex, and race of the defendant; number of charges in the case; whether the defendant pled guilty; if the case included a weapon enhancement; the sentencing guidelines in place at the time of the case; and the type of offense.

Table A.8: Linear Regression Results from Pooled Dataset

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	-0.099 (0.120)	0.025 (0.077)	-0.058 (0.120)	0.075 (0.079)	-0.064 (0.119)	0.046 (0.077)
Guidelines Upper	0.021*** (0.003)	0.015*** (0.002)	0.021*** (0.003)	0.015*** (0.002)	0.020*** (0.003)	0.014*** (0.002)
Guidelines Lower	0.094*** (0.004)	0.068*** (0.004)	0.094*** (0.004)	0.068*** (0.004)	0.095*** (0.004)	0.068*** (0.004)
Male	1.851*** (0.080)	1.476*** (0.063)	1.858*** (0.080)	1.480*** (0.063)	1.851*** (0.080)	1.476*** (0.063)
Non-White	0.183** (0.069)	0.452*** (0.038)	0.190** (0.070)	0.456*** (0.039)	0.185** (0.070)	0.454*** (0.038)
Age	0.088*** (0.015)	0.073*** (0.011)	0.090*** (0.016)	0.073*** (0.011)	0.088*** (0.015)	0.073*** (0.011)
Squared Age	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Plea	-0.998*** (0.126)	-0.779*** (0.078)	-0.959*** (0.125)	-0.733*** (0.079)	-1.001*** (0.124)	-0.765*** (0.079)
Weapon Used	0.211 (0.142)	0.237* (0.108)	0.221 (0.141)	0.245* (0.108)	0.204 (0.142)	0.232* (0.108)
Weapon Enhancement	1.236*** (0.137)	0.813*** (0.091)	1.238*** (0.138)	0.818*** (0.093)	1.255*** (0.137)	0.825*** (0.092)
6th Ed.	-0.656*** (0.142)	-0.592*** (0.110)	0.081 (0.116)	0.010 (0.088)	-0.387** (0.129)	-0.340*** (0.096)
7th Ed.	-1.073*** (0.207)	-0.842*** (0.158)	0.093 (0.138)	-0.016 (0.094)	-0.879*** (0.209)	-0.739*** (0.148)
No. of Charges	0.027 (0.022)	0.020 (0.017)	0.028 (0.022)	0.020 (0.018)	0.027 (0.022)	0.020 (0.017)
Rape	2.765*** (0.221)	1.676*** (0.166)	2.874*** (0.222)	1.761*** (0.167)	2.795*** (0.218)	1.700*** (0.165)
Robbery	0.767*** (0.060)	0.646*** (0.046)	0.754*** (0.060)	0.637*** (0.045)	0.761*** (0.059)	0.641*** (0.045)
Num.Obs.	47 297	47 297	47 297	47 297	47 297	47 297
R2	0.553	0.544	0.551	0.542	0.552	0.543
Box-Cox λ	0.500	0.400	0.500	0.400	0.500	0.400

Table A.9: Non-Linear Box-Cox Models

The Box-Cox transformation addresses potential non-linearities and non-normal error distributions by defining a flexible functional form for the dependent variable, $Y^{(\lambda)} = \frac{Y^\lambda - 1}{\lambda}$ when $\lambda \neq 0$, and $Y^{(\lambda)} = \ln(Y)$ when $\lambda = 0$. Because minimum sentence lengths include values of zero months and the transformation strictly requires $Y > 0$, we apply a standard scaling shift of +0.01 months to the outcome variable prior to estimation. For each specification, the parameter λ is estimated independently via maximum likelihood estimation over a search grid of $[-2, 2]$. The optimal λ values chosen to maximize normality and homoscedasticity are reported at the bottom of the table.

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	0.918 (2.493)	-0.258 (2.461)	1.263 (2.758)	-1.090 (2.537)	0.730 (2.494)	-0.748 (2.491)
Min. Sentence	-0.725*** (0.040)	-0.733*** (0.041)	-0.736*** (0.040)	-0.734*** (0.041)	-0.727*** (0.040)	-0.732*** (0.041)
Max. Sentence	1.378*** (0.047)	1.389*** (0.048)	1.399*** (0.047)	1.393*** (0.048)	1.387*** (0.047)	1.392*** (0.048)
Male	28.210*** (1.527)	27.738*** (1.566)	30.241*** (1.529)	28.613*** (1.538)	28.315*** (1.523)	27.788*** (1.563)
Non-White	-1.387 (1.385)	-0.408 (1.581)	0.569 (1.509)	0.435 (1.589)	-1.716 (1.383)	-0.497 (1.577)
Age	4.391*** (0.411)	4.394*** (0.421)	3.859*** (0.408)	4.224*** (0.421)	4.447*** (0.416)	4.418*** (0.423)
Age Squared	-0.053*** (0.005)	-0.053*** (0.006)	-0.047*** (0.005)	-0.051*** (0.006)	-0.053*** (0.006)	-0.053*** (0.006)
Guilty Plea	-65.441*** (3.169)	-64.884*** (3.215)	-68.306*** (3.009)	-64.274*** (3.119)	-61.505*** (3.105)	-61.719*** (3.135)
Total Offenses	2.845*** (0.152)	2.939*** (0.152)	2.736*** (0.145)	3.040*** (0.146)	3.128*** (0.149)	3.172*** (0.149)
Linear Time					-49.665*** (3.510)	-51.729*** (5.047)
Num.Obs.	31 474	31 474	31 474	31 474	31 474	31 474
R2	0.696	0.704	0.689	0.700	0.694	0.702
Time Control	Year FE	Year FE	No	No	Linear	Linear
Judge FE	No	Yes	No	Yes	No	Yes

*** p < 0.01

Robust standard errors in parentheses are clustered at the judge level. Proximity is a linear measure of time until the next retention election. A proximity of 0 is equivalent to the first day of a judge's term, while a proximity of 1 is the last day of the term, ten years later. A positive coefficient would be consistent with the story of election cycles in sentencing. "Linear" time control denotes the temporal position of each observation within the dataset, capturing how far into the span of the data, from 2000 to 2024, the observation occurs. There is no covariate for guidelines edition as there was no change in the guidelines during the period studied. There is no covariate for weapons enhancements as this data was not provided. Data is courtesy of the Colorado State Court Administrator's Office.

Table A.10: Linear Regression Results from Colorado

	(1)	(2)	(3)	(4)	(5)	(6)
Proximity	-0.576 (0.570)	-0.064 (0.501)	-0.439 (0.564)	0.132 (0.513)	-0.456 (0.560)	0.039 (0.506)
Guidelines Upper	0.133*** (0.021)	0.136*** (0.020)	0.132*** (0.021)	0.134*** (0.020)	0.131*** (0.021)	0.133*** (0.020)
Guidelines Lower	0.721*** (0.023)	0.731*** (0.022)	0.723*** (0.023)	0.734*** (0.022)	0.724*** (0.023)	0.734*** (0.022)
Male	4.592*** (0.315)	4.132*** (0.293)	4.611*** (0.319)	4.144*** (0.295)	4.593*** (0.317)	4.132*** (0.293)
Non-White	-0.446 (0.326)	1.118*** (0.222)	-0.405 (0.332)	1.115*** (0.222)	-0.419 (0.330)	1.108*** (0.221)
Age	0.093 (0.064)	0.078 (0.059)	0.093 (0.064)	0.075 (0.059)	0.089 (0.064)	0.072 (0.059)
Squared Age	-0.002** (0.001)	-0.002* (0.001)	-0.002** (0.001)	-0.002* (0.001)	-0.002** (0.001)	-0.002* (0.001)
Plea	-7.013*** (0.664)	-7.674*** (0.570)	-6.698*** (0.683)	-7.231*** (0.596)	-6.804*** (0.674)	-7.338*** (0.594)
Weapon Used	-0.298 (0.722)	0.275 (0.694)	-0.262 (0.721)	0.326 (0.697)	-0.304 (0.725)	0.284 (0.698)
Weapon Enhancement	1.038 (0.784)	-0.365 (0.631)	1.001 (0.786)	-0.389 (0.639)	1.042 (0.788)	-0.364 (0.641)
6th Ed.	-2.391** (0.816)	-2.947*** (0.778)	0.217 (0.526)	0.032 (0.489)	-0.937 (0.646)	-1.114 (0.619)
7th Ed.	-3.596** (1.185)	-3.880** (1.181)	-0.408 (0.650)	-0.763 (0.565)	-2.806** (1.032)	-3.136** (1.016)
No. of Charges	0.162 (0.090)	0.174 (0.102)	0.165 (0.092)	0.178 (0.103)	0.163 (0.091)	0.176 (0.102)
Rape	15.022*** (1.300)	13.088*** (1.235)	15.362*** (1.305)	13.474*** (1.248)	15.167*** (1.280)	13.272*** (1.232)
Robbery	3.453*** (0.274)	3.645*** (0.238)	3.430*** (0.272)	3.634*** (0.237)	3.447*** (0.272)	3.647*** (0.237)
Num.Obs.	47 297	47 297	47 297	47 297	47 297	47 297
R2	0.689	0.714	0.688	0.713	0.688	0.713
R2 Adj.	0.689	0.711	0.688	0.710	0.688	0.711
RMSE	19.88	19.06	19.89	19.08	19.89	19.08
Time Control	Year FE	Year FE	No	No	Linear	Linear
Judge FE	No	Yes	No	Yes	No	Yes

* p < 0.1, ** p < 0.05, *** p < 0.01.

Robust standard errors in parentheses are clustered at the judge level. Proximity is a linear measure of time until the next retention election. A proximity of 0 is equivalent to the first day of a judge's term, while a proximity of 1 is the last day of the term, ten years later. A positive coefficient would be consistent with the story of election cycles in sentencing. "Linear" time control denotes the temporal position of each observation within the dataset, capturing how far into the 20-year span of 2001-2021 the observation occurs.

Table B.1: Linear Regression Results with Full Set of Covariates Reported

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Proximity	0.052 (0.411)	-0.171 (0.500)	-0.018 (0.371)	0.021 (0.341)	-0.058 (0.036)	0.034 (0.436)	0.062 (0.276)
County Unemployment	0.622*** (0.087)	0.599*** (0.105)	0.218** (0.077)	0.502*** (0.072)	0.016* (0.007)	0.697*** (0.099)	0.562*** (0.059)
Violent Crime Rate						-0.034 (0.251)	
Total Crime Rate						0.049 (0.039)	
Defendant Age	0.011 (0.013)	-0.020 (0.015)	-0.019 (0.011)	-0.001 (0.010)	-0.007*** (0.001)	0.010 (0.014)	0.015 (0.009)
Plea Deal	-11.379*** (0.452)	-11.474*** (0.503)		-7.497*** (0.313)	-0.223*** (0.029)	-11.089*** (0.459)	-9.383*** (0.312)
Guidelines Lower Bound				0.815*** (0.018)			
Guidelines Upper Bound				0.103*** (0.015)			
Num.Obs.	47 297	52 002	38 942	47 297	47 297	43 966	125 110
R2	0.550	0.947	0.575	0.716	0.345	0.542	0.917
R2 Adj.	0.544	0.947	0.569	0.713	0.337	0.537	0.917
RMSE	23.91	29.18	18.94	18.98	2.07	24.08	25.49
Case Selection	H&G	B&Y	H&G	H&G	H&G	H&G	Severity
Adjudication	All	All	Pleas	All	All	All	All
Outcome	Topcoded	Topcoded	Topcoded	Log	Topcoded	Topcoded	Topcoded
Guidelines	Cell	Cell	Cell	Cell	Upper/Lower	Cell	Cell
Crime Rates	N	N	N	N	N	Y	N
County FE	Y	Y	Y	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y	Y	Y	Y
Offense FE	Y	Y	Y	Y	Y	Y	Y

* p < 0.1, ** p < 0.05, *** p < 0.01.

Robust standard errors in parentheses are clustered at the judge level. Proximity is a linear measure of time until the next retention election. A proximity of 0 is equivalent to the first day of a judge's term, while a proximity of 1 is the last day of the term, ten years later. For case selection, "H&G" includes offenses from the [Huber and Gordon \(2004\)](#) article (aggravated assault, robbery, and rape), while "B&Y" includes additional offenses specified by [Berdejó and Yuchtman \(2013\)](#) (murder, manslaughter). "Severity" includes all class 1 and 2 felonies along with murder and manslaughter. For adjudication, "all" refers to cases which went to trial or resulted in a guilty plea, while "pleas" excludes trials. For outcome, "topcoded" censors the sentence at 720 months, while "log" is a log of the topcoded outcome. For guidelines, "cell" indicates a factor control variable for the cell of the sentencing guidelines, while "upper/lower" includes separate numeric controls for the upper and lower suggested sentence for the cell. Crime rates indicates the presence of controls for county level violent and total crime rates.

Table B.2: Additional Model Specifications With Full Set of Covariates Reported